AMATH 403 HW 7

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Question 1: 8.1.4

(a) Use the Fourier transform to solve the initial value problem

$$\frac{\partial u}{\partial t} = \frac{\partial^2 u}{\partial x^2}, \qquad u(0,x) = \delta'(x-\xi), \qquad -\infty < x < \infty, \quad t > 0,$$

whose initial data is the derivative of the delta function at a fixed position ξ.

(b) Show that your solution can be written as the derivative $\partial F/\partial x$ of the fundamental solution $F(t,x;\xi)$. Explain why this observation should be valid.